

GTPO: GROUP TEACHING POLICY OPTIMIZATION VIA TEMPORAL OPTIMAL TRANSPORT IN VLA REINFORCEMENT LEARNING

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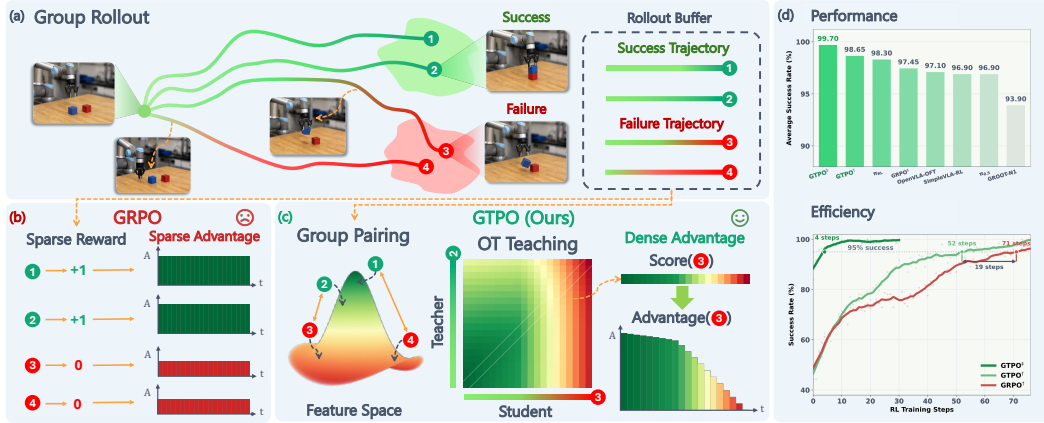


Figure 1: **GTPO overview.** (a) From the same initial environment state, the policy samples a group of successful and failed trajectories and stores them in the rollout buffer. (b) GRPO broadcasts each sparse trajectory-level advantage uniformly across all action chunks. (c) GTPO pairs each failure with its nearest successful rollout in visual feature space and uses temporal OT alignment to produce a dense advantage over time. (d) This dense credit assignment improves both average success rate and RL training efficiency under the same rollout budget and policy initialization.

ABSTRACT

Reinforcement learning (RL) is becoming increasingly important for post-training robot policies: by continually interacting with the environment, a policy can improve beyond the ceiling imposed by behavior cloning. However, robot interaction is expensive, and training often relies on sparse terminal outcomes that provide little information about which actions caused success or failure, leading to inefficient or unstable optimization. Obtaining dense process-level supervision is therefore a central challenge. Inspired by inverse reinforcement learning and built on Group Relative Policy Optimization (GRPO), we propose Group Teaching Policy Optimization (GTPO). Within each group of rollouts from the same initial configuration, GTPO represents every rollout as a trajectory of visual features, pairs each failure with its nearest successful rollout as an on-policy teacher, and uses temporal optimal transport to convert the sparse trajectory-level signal into dense chunk-level credit. A temporal mask restricts distant matches during alignment, while the in-group teacher improves together with the policy. OT teaching requires no additional expert reference trajectories or reward model. On all four LIBERO suites, GTPO reaches 98.65% average success from a one-demonstration-per-task warm start and 99.70% with the full demonstration set, consistently outperforming GRPO, particularly on long-horizon tasks. We further evaluate GTPO on five real-world manipulation tasks of increasing difficulty to examine whether its sample-efficiency gains transfer from simulation to physical interaction under realistic deployment variation and imperfect object placement encountered during physical execution.

1 INTRODUCTION

Vision-language-action (VLA) models provide a general interface from observations and language instructions to robot actions (Kim et al., 2024), but complex, fine-grained manipulation still requires task-specific post-training. Reinforcement learning (RL) is particularly attractive because interaction can improve a policy beyond its demonstrations (Hu et al., 2025; Tan et al., 2025; Li et al., 2026; Zang et al., 2025). In robotics, however, interaction is costly and feedback is usually limited to a sparse terminal outcome. Success or failure alone cannot reveal which grasp, alignment, or placement decision determined the result, leaving long-horizon optimization inefficient and unstable. The central challenge is therefore to obtain reliable process-level credit without dense manual annotation or task-specific reward engineering.

Our motivation comes from combining two complementary ideas. Inverse RL and related trajectory-matching work show that successful behavior can reveal richer supervision than a terminal outcome alone (Ng & Russell, 2000; Abbeel & Ng, 2004; Zakka et al., 2021; Haldar et al., 2023; Luo et al., 2023; Fu et al., 2024), although they commonly obtain this signal from fixed expert trajectories and formulate an additional reward. GRPO (Shao et al., 2024), meanwhile, samples groups of on-policy rollouts from the same initial state and estimates their relative advantages without a critic. When a group contains both successes and failures, the successful rollouts provide the behavioral references suggested by inverse RL, but now they are generated by the current policy and improve along with it. Standard GRPO does not exploit this opportunity: it reduces every trajectory to one scalar advantage and broadcasts that value uniformly across all action chunks, leaving temporal differences between successful and failed behavior unused.

Motivated by this observation, we propose **Group Teaching Policy Optimization (GTPO)**. Within each rollout group, GTPO treats successful members as on-policy teachers and pairs each failure with its nearest success in visual trajectory space. Temporally masked Sinkhorn OT aligns the pair and redistributes the original trajectory-level advantage into chunk-level credit. OT therefore acts as a *credit allocator*, rather than a new reward: its teachers come from the policy’s own rollouts, requiring no additional expert reference trajectories or reward model. The terminal reward is retained while temporal updates are reweighted; SFT initialization and real-world corrective data remain part of training.

Our contributions are: (1) an on-policy group-teaching framework that turns successful group members into adaptive teachers; (2) a temporally masked OT mechanism that decomposes scalar GRPO advantages in the VLA’s native feature space; and (3) an evaluation spanning four LIBERO suites and five real-world tasks of increasing difficulty, testing both learning efficiency and transfer to fine-grained physical interaction. On LIBERO, GTPO reaches 98.65% average success from one demonstration per task and 99.70% with the full set, with the largest gains on long-horizon tasks where delayed feedback makes temporal credit assignment especially challenging.

2 RELATED WORK

2.1 REINFORCEMENT LEARNING FOR ROBOT MANIPULATION

Robot RL combines demonstrations with online interaction to improve sample efficiency. RLPD mixes prior data with online replay (Ball et al., 2023); SERL adds learned rewards and automated resets (Luo et al., 2024a); and HIL-SERL uses human interventions for contact-rich manipulation (Luo et al., 2024b). RLDG distills specialist experience into generalist policies (Xu et al., 2025), while FLaRe scales RL fine-tuning to pretrained transformers (Hu et al., 2025). Together, these systems show that interaction can refine pretrained visuomotor representations while preserving the observation and action interfaces required for scalable VLA post-training.

RL has also become a post-training stage for VLA models. GRAPE learns from stage-wise preferences (Zhang et al., 2024); RIPT-VLA combines dynamic sampling with sparse binary feedback (Tan et al., 2025); SimpleVLA-RL studies scalable online optimization across manipulation benchmarks (Li et al., 2026); and RLinf-VLA provides unified PPO/GRPO infrastructure across VLAs and simulators (Zang et al., 2025). These works demonstrate the value of interaction, whereas GTPO addresses the complementary problem of temporal credit assignment in critic-free group-based RL. In particular, sparse outcome-based methods still assign the same rollout-level signal to every

action, while critic-based alternatives require learning an additional value function from limited robot interaction. GTPO retains GRPO’s critic-free optimization but introduces structure within each trajectory by comparing rollouts generated under matched initial conditions, reducing variation due to scene configuration.

2.2 DENSE REWARD METHODS IN ROBOTIC REINFORCEMENT LEARNING

Dense reward learning provides intermediate supervision by estimating task progress from richer signals. REDS uses segmented videos and subtask conditioning (Kim et al., 2025); ReWiND models progress and regression from language and rewound demonstrations (Zhang et al., 2025); and DEMO³ jointly learns multi-stage rewards, policies, and world models (López Escoriza et al., 2025). For real-robot learning, VLAC predicts progress from paired observations (Zhai et al., 2025). Other methods exploit robot corpora, procedural descriptions, counterfactual failures, or learned dynamics (Lee et al., 2026; Wu et al., 2026; Feng et al., 2026; Chen et al., 2026; Lv et al., 2026; Fang et al., 2026). RoboReward, for example, uses large robot corpora and counterfactual negatives, whereas procedural reward models ground feedback in predicted task stages. Dynamics-aware approaches additionally estimate whether an observation represents physically plausible progress. Although these methods can produce detailed process supervision, they require a learned or prompted reward mechanism whose errors may alter the optimization objective. GTPO trains no reward model: it retains the terminal task reward and uses successful group members to reweight its advantage across time. This changes the policy-update weighting without introducing a separately learned progress reward.

2.3 INVERSE REINFORCEMENT LEARNING AND TEMPORAL ALIGNMENT

IRL recovers a reward or progress signal from expert behavior when task rewards are unavailable or sparse. XIRL learns visual progress across embodiments (Zakka et al., 2021), while OT-based imitation aligns learner and expert trajectories or occupancy measures (Haldar et al., 2023; Luo et al., 2023). TemporalOT (Fu et al., 2024) introduces temporal locality into OT and converts similarity to a fixed expert demonstration into per-step rewards. GTPO transfers this principle to online group-based RL but differs in two respects: the nearest successful rollout in the current group acts as an adaptive teacher, and OT allocates an existing GRPO advantage instead of defining a new reward. The teacher thus improves together with the policy and needs no additional expert trajectory for OT teaching. Because unconstrained OT is insensitive to sequence order, a temporal support mask restricts matches between distant manipulation phases; it does not establish causal correspondence or enforce strict monotonicity.

3 METHOD

3.1 PROBLEM FORMULATION

Given the current observation o_t and a language instruction ℓ , a policy π_θ predicts an action chunk

$$a_t = (a_t^0, \dots, a_t^{n-1}) \sim \pi_\theta(\cdot \mid o_t, \ell), \quad (1)$$

where n is the chunk size. The robot executes a_t , interacts with the environment, and receives the next observation o_{t+1} until reaching the goal or horizon T .

Both successful and failed trajectories are stored in the rollout buffer. A successful trajectory that reaches the goal at step $g \leq T$ and a failed trajectory that exhausts the horizon are, respectively,

$$\zeta^{\text{succ}} = (o_0, a_0, o_1, a_1, \dots, o_{g-1}, a_{g-1}, o_g), \quad \zeta^{\text{fail}} = (o_0, a_0, o_1, a_1, \dots, o_{T-1}, a_{T-1}, o_T). \quad (2)$$

Each trajectory receives only a terminal reward indicating goal completion or horizon exhaustion,

$$r(\zeta) = \begin{cases} r_{\text{succ}} > 0, & \text{goal reached within } T, \\ r_{\text{fail}} < 0, & \text{otherwise.} \end{cases} \quad (3)$$

Training maximizes the expected terminal reward over tasks and policy-generated trajectories:

$$\theta^* = \underset{\theta}{\operatorname{argmax}} J(\theta), \quad J(\theta) = \mathbb{E}_{\ell, \zeta \sim \pi_\theta} [r(\zeta)]. \quad (4)$$

3.2 PRELIMINARIES: GROUP RELATIVE POLICY OPTIMIZATION

GRPO (Shao et al., 2024) avoids a value model by sampling G rollouts from the same initial state and normalizing their terminal rewards within the group:

$$A_i = \frac{r_i - \mu_g}{\sigma_g + \varepsilon_{\text{std}}}, \quad \mu_g = \frac{1}{G} \sum_{j=1}^G r_j, \quad \sigma_g = \sqrt{\frac{1}{G-1} \sum_{j=1}^G (r_j - \mu_g)^2}, \quad (5)$$

where $\varepsilon_{\text{std}} > 0$. With importance ratio $\rho_{i,t} = \pi_\theta(a_{i,t} \mid o_{i,t}, \ell) / \pi_{\theta_{\text{old}}}(a_{i,t} \mid o_{i,t}, \ell)$, its clipped objective is

$$\mathcal{L}_{\text{GRPO}}(\theta) = - \frac{1}{\sum_{i,t} m_{i,t}} \sum_{i=1}^G \sum_{t=0}^{T_i-1} m_{i,t} \min(\rho_{i,t} A_i, \text{clip}(\rho_{i,t}, 1 - \varepsilon_{\text{clip}}, 1 + \varepsilon_{\text{clip}}) A_i), \quad (6)$$

where T_i is the rollout length and $m_{i,t}$ masks padding. The same A_i is broadcast to every chunk, providing no within-trajectory credit variation (Figure 1(b)). GTPO instead derives a temporally shaped $\hat{A}_{i,t}$ from same-state grouped rollouts while retaining this critic-free clipped-loss form.

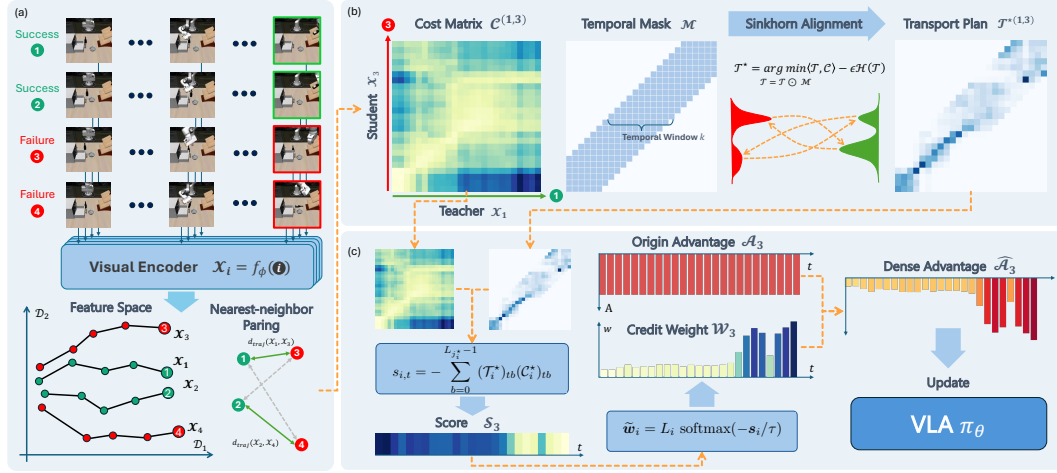


Figure 2: **GTPO pipeline.** (a) Rollouts are encoded as visual feature trajectories, and each failed rollout is paired with its nearest successful teacher. (b) A temporally masked Sinkhorn plan aligns the paired trajectories. (c) The alignment produces chunk-level credit weights that reshape the original GRPO advantage for policy optimization without altering the rollout collection procedure.

3.3 NEAREST-NEIGHBOR GROUP TEACHING

We encode each rollout, select its nearest in-group success as teacher, and align the pair with temporal OT (Figure 2(a)).

Visual features and pairwise cost. For T_i valid action chunks, the policy encoder f_ϕ supplies the shifted feature sequence

$$x_{i,t} = f_\phi(o_{i,t}), \quad \mathcal{X}_i = (x_{i,1}, \dots, x_{i,T_i-1}) \in \mathbb{R}^{L_i \times D}, \quad L_i = T_i - 1. \quad (7)$$

Here $o_{i,t+1}$ follows $a_{i,t}$; the final action has no stored successor feature and retains weight 1. Let $\mathcal{S} = \{j : r_j > 0\}$ and $\mathcal{F} = \{i : r_i < 0\}$ denote successes and failures. For each failed–successful pair $(i, j) \in \mathcal{F} \times \mathcal{S}$, cosine distance gives

$$\mathcal{C}_{ab}^{(i,j)} = 1 - \frac{\langle x_{i,a+1}, x_{j,b+1} \rangle}{\|x_{i,a+1}\| \|x_{j,b+1}\| + \delta}, \quad (8)$$

where $0 \leq a < L_i$, $0 \leq b < L_j$, and $\delta > 0$ ensures numerical stability.

Temporally masked Sinkhorn alignment. Following TemporalOT (Fu et al., 2024), we restrict entropic Sinkhorn alignment (Peyré & Cuturi, 2019; Cuturi, 2013) to a temporal band:

$$\mathcal{M}_{ab}^{(i,j)} = \begin{cases} 1, & L_i = L_j \text{ and } |a - b| \leq k, \\ 1, & L_i \neq L_j \text{ and } \left| \frac{a}{\max(L_i-1,1)} - \frac{b}{\max(L_j-1,1)} \right| \leq \frac{k}{\max(L_i, L_j)}, \\ 0, & \text{otherwise,} \end{cases} \quad (9)$$

where k controls temporal tolerance. With uniform marginals $\mu = \frac{1}{L_i} \mathbf{1}$ and $\nu = \frac{1}{L_j} \mathbf{1}$, the temporally masked transport plan is

$$\begin{aligned} \mathcal{T}^{*(i,j)} &= \underset{\substack{\mathcal{T} \geq 0 \\ \mathcal{T} = \mathcal{T} \odot \mathcal{M}^{(i,j)}}}{\operatorname{argmin}} \quad \langle \mathcal{T}, \mathcal{C}^{(i,j)} \rangle - \varepsilon H(\mathcal{T}) \\ \text{s.t.} \quad \mathcal{T} \mathbf{1} &= \frac{1}{L_i} \mathbf{1}, \quad \mathcal{T}^\top \mathbf{1} = \frac{1}{L_j} \mathbf{1}. \end{aligned} \quad (10)$$

Here $\mathcal{T}_{ab}$ is transported mass, $H(\mathcal{T}) = -\sum_{a,b} \mathcal{T}_{ab} \log \mathcal{T}_{ab}$, and the mask zeros matches outside the band. Appendix A.2 derives the solution. Total transport cost defines the distance between shifted rollout sequences:

$$d_{\text{traj}}(\mathcal{X}_i, \mathcal{X}_j) = \langle \mathcal{T}^{*(i,j)}, \mathcal{C}^{(i,j)} \rangle. \quad (11)$$

Nearest-neighbor teacher. Each failure selects the successful teacher with minimum transport cost:

$$j_i^* = \underset{j \in \mathcal{S}}{\operatorname{argmin}} d_{\text{traj}}(\mathcal{X}_i, \mathcal{X}_j), \quad i \in \mathcal{F}. \quad (12)$$

Pairs with $T_i < 2$ or $T_j < 2$ are not aligned. If no valid teacher is available, weights remain uniform, reducing the update to the original GRPO assignment.

3.4 CHUNK-LEVEL CREDIT ASSIGNMENT

For the selected teacher, let $\mathcal{T}_i^* = \mathcal{T}^{*(i,j_i^*)}$ and $\mathcal{C}_i^* = \mathcal{C}^{(i,j_i^*)}$. The per-chunk alignment scores are

$$s_{i,t} = - \sum_{b=0}^{L_{j_i^*}-1} (\mathcal{T}_i^*)_{tb} (\mathcal{C}_i^*)_{tb}, \quad 0 \leq t < L_i. \quad (13)$$

Negating these alignment scores yields penalty weights,

$$\tilde{w}_{i,t} = L_i \frac{\exp(-s_{i,t}/\tau)}{\sum_{u=0}^{L_i-1} \exp(-s_{i,u}/\tau)}, \quad \sum_{t=0}^{L_i-1} \tilde{w}_{i,t} = L_i. \quad (14)$$

Higher cost means lower $s_{i,t}$ and larger $\tilde{w}_{i,t}$, amplifying the magnitude of negative advantage on more divergent chunks. Temperature τ controls concentration. The shifted features already associate row t with action $a_{i,t}$: set $w_{i,t} = \tilde{w}_{i,t}$ for $t < L_i$ and $w_{i,T_i-1} = 1$, giving $\sum_t w_{i,t} = T_i$. Rollouts with $T_i < 2$ retain uniform weights.

$$\hat{A}_{i,t} = A_i \cdot w_{i,t} \cdot m_{i,t}. \quad (15)$$

The resulting dense advantage $\hat{A}_{i,t}$ directly replaces A_i in the GRPO loss (6), preserving its clipped-loss form while changing the temporal weighting of policy updates.

Algorithm 1 gives the full procedure. Defaults are NNT assignment, $G = 8$, $k = 30$, $\varepsilon = 0.01$ (100 iterations), cosine distance, and exponential $\tau : 0.05 \rightarrow 1$; Section 4.4 ablates these choices.

Properties of the credit transformation. Since $\sum_t w_{i,t} = T_i$, the mean chunk advantage remains A_i before policy ratios and clipping. This normalization does not preserve the loss, gradient, or KL change of a GRPO update. The temporal mask restricts temporally distant matches, while nearest-neighbor selection adapts the reference to each failure. Successful rollouts and groups without a success retain uniform weights, so temporal shaping is applied only when the group provides direct evidence of better behavior.

4 SIMULATION EXPERIMENTS

Our experiments test whether GTPO improves over GRPO and prior VLA RL methods, whether the gain is demonstration-efficient, and which design choices (temporal bandwidth, OT temperature and schedule, distance metric, and teacher assignment) drive the improvement across suites with different task structures and horizons represented by the four LIBERO suites.

4.1 SETUP

We report pass@1 success on all four LIBERO (Liu et al., 2023) suites using one third-person camera. GRPO and GTPO share the same LoRA-adapted OpenVLA-OFT (Kim et al., 2024) warm start, using either one expert trajectory per task ($\dagger$) or the full demonstrations ($\ddagger$); these checkpoints are weaker than the published full model. Training uses RLinf (Yu et al., 2025) with $G = 8$, terminal binary rewards, and token-level optimization; trajectory advantages are broadcast to valid chunks before temporal reweighting. Published baselines ($\S$) are included as references under their original settings. This protocol isolates post-training gains while holding the policy backbone, rollout grouping, and optimization interface fixed throughout the matched comparison.

4.2 MAIN RESULTS

Table 1: LIBERO success rate. $\dagger/\ddagger$: one/full demonstration warm start; $\S$: published result under its original setting. Parentheses show gains over the matched SFT checkpoint.

Models	Paradigm	Object		Spatial		Goal		Long		Average	
		SR $\uparrow$	Rank $\downarrow$	SR $\uparrow$	Rank $\downarrow$	SR $\uparrow$	Rank $\downarrow$	SR $\uparrow$	Rank $\downarrow$	SR $\uparrow$	Rank $\downarrow$
OpenVLA $\S$	SFT	88.4	13	84.7	14	79.2	15	53.7	15	76.5	14
GROOT-N1 $\S$	SFT	97.6	9	94.4	10	93.0	11	90.6	10	93.9	10
$\pi_0^\S$	SFT	98.8	4	96.8	8	95.8	9	85.2	11	94.2	9
$\pi_{0.5}^\S$	SFT	98.2	7	98.8	4	98.0	7	92.4	7	96.9	7
OpenVLA-OFT $\S$	SFT	98.4	6	97.6	7	97.9	8	94.5	3	97.1	6
GRAPE $\S$	RL	92.1	11	88.5	13	83.1	12	57.2	14	80.2	13
TGRPO $\S$	RL	92.2	10	90.4	11	81.0	14	59.2	13	80.7	12
VLA-RL $\S$	RL	91.8	12	90.2	12	82.2	13	59.8	12	81.0	11
SimpleVLA-RL $\S$	RL	98.7	5	98.2	6	98.8	4	91.7	8	96.9	7
$\pi_{RL}^\S$	RL	100.0	1	99.6	2	99.6	2	94.0	4	98.3	3
OpenVLA-OFT $\dagger$	SFT	28.8	15	52.25	15	49.4	16	14.9	16	36.34	16
GRPO $\dagger$	RL	97.7	8	94.8	9	94.0	10	91.0	9	94.38	8
GTPO $\dagger$ (Ours)	RL	100.0 (+71.2)	1	98.6 (+46.35)	5	99.2 (+49.8)	3	96.8 (+81.9)	2	98.65 (+62.31)	2
OpenVLA-OFT $\ddagger$	SFT	50.2	14	51.6	16	49.4	16	11.9	17	40.78	15
GRPO $\ddagger$	RL	99.6	3	98.6	5	98.1	6	93.5	6	97.45	5
GTPO $\ddagger$ (Ours)	RL	100.0 (+49.8)	1	100.0 (+48.4)	1	99.8 (+50.4)	1	99.0 (+87.1)	1	99.70 (+58.93)	1

Table 2: Success rate (%) on four RoboTwin fine-grained manipulation tasks.

Method	Place Container Plate	Place Empty Cup	Beat Block Hammer	Pick Dual Bottles	Average
GRPO	95.3	94.5	96.1	92.3	94.55
GTPO (Ours)	99.2	98.5	98.7	98.1	98.63

(1) Strong performance with limited demonstrations. With only one demonstration per task, GTPO reaches 98.65% average success, already exceeding full-data GRPO (97.45%). Using the full demonstration set further raises performance to 99.70%, ranking first on all four LIBERO suites. Notably, the gap between the one- and full-demonstration settings is only 1.05 points for GTPO, compared with 3.07 points for GRPO, indicating substantially reduced dependence on large demonstration sets.

(2) Consistent gains from temporal group teaching. Compared with matched GRPO, GTPO improves every suite under both warm-start settings, with average gains of 4.27 and 2.25 points. The largest margins occur on LIBERO-Long (+5.8/ +5.5), showing that temporally resolved credit is particularly valuable when sparse outcomes must be propagated through long action sequences.

The persistent long-horizon advantage across both warm starts suggests that the gain is not merely compensation for a weak SFT model.

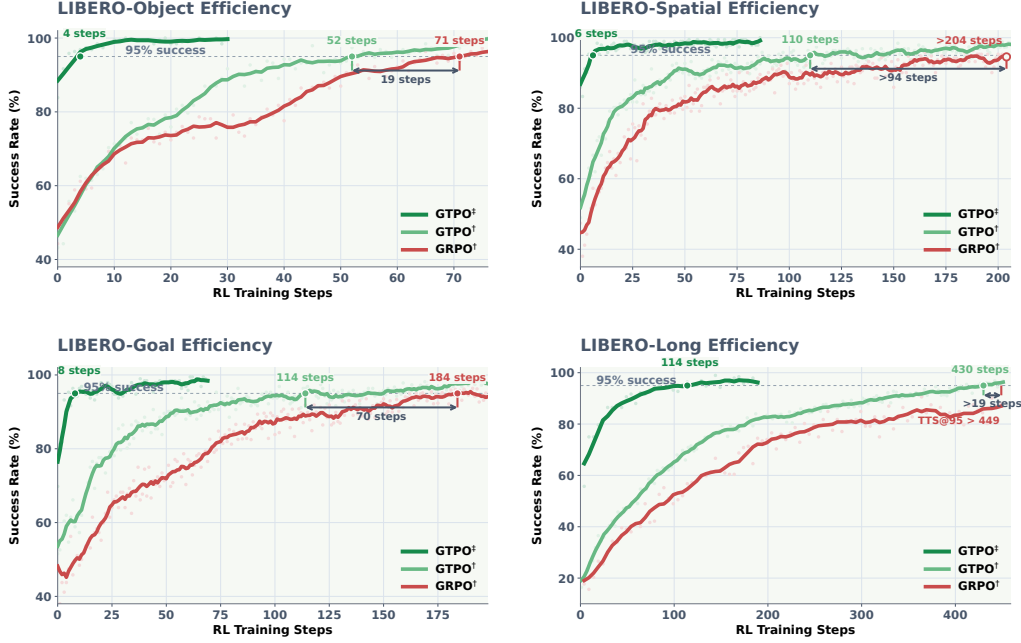


Figure 3: Training efficiency on LIBERO-Object (top left) and LIBERO-Spatial (top right), LIBERO-Goal (bottom left), and LIBERO-Long (bottom right). Curves show seven-checkpoint moving averages of success rate, with raw evaluations shown as pale points.

Training efficiency. We summarize Figure 3 using normalized AUC-SR, success at 25% of training (Early SR), and steps to 95% success (TTS@95), macro-averaged across the four suites. These metrics capture full-horizon sample efficiency, early learning progress, and the time needed to obtain a reliable policy, respectively.

Table 3: Aggregate training efficiency on LIBERO. For TTS@95, runs that do not reach 95% are censored at their training budget, making the reported improvement a conservative lower bound.

Method	AUC-SR $\uparrow$	Early SR@25% $\uparrow$	TTS@95 $\downarrow$
GRPO †	78.55	70.46	> 227.0
GTPO †	85.92	81.63	176.5
Improvement	+7.37	+11.17	> 22.2% fewer steps

(3) Faster optimization and consistent cross-benchmark gains. GTPO gains 7.37 AUC-SR and 11.17 Early SR while requiring at least 22.2% fewer steps to reach 95% success. The curves therefore show an advantage throughout training rather than only at the final checkpoint. GTPO also improves every RoboTwin task, raising average success from 94.55% to 98.63%. The consistent ordering across LIBERO and fine-grained RoboTwin tasks indicates that the gain is not specific to a single benchmark or evaluation granularity.

4.3 OT NECESSITY ABLATION

We compare GTPO with NNT + pointwise cosine matching (GTPO-pointwise) on LIBERO-Object. Pointwise matching replaces masked Sinkhorn while retaining nearest-success teaching and normalized chunk weights; Table 4 summarizes the three methods. GTPO exceeds pointwise matching by 4.08 AUC-SR points and reaches 95% success 21 steps earlier (Figure 4), supporting this comparison but not establishing universal OT necessity or causal localization.

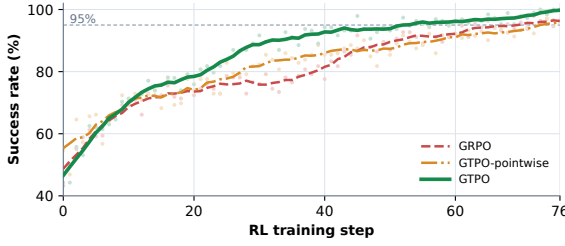


Figure 4: OT necessity ablation on LIBERO-Object. Curves show the recorded success rates for GRPO, GTPO-pointwise, and GTPO. Lines are seven-checkpoint moving averages; pale points are raw evaluations, not seed-level uncertainty.

Table 4: OT-versus-pointwise matching on LIBERO-Object. Metrics use the recorded success rates. Final SR is the final raw evaluation; other metrics use seven-checkpoint moving averages. Appendix B.4 gives definitions.

Method	AUC-SR	Early SR	TTS@95	Final SR
GRPO	80.72	73.74	70	96.53
GTPO-pointwise	81.89	74.70	73	95.80
GTPO	85.97	78.23	52	100.00

4.4 ANALYSIS OF GROUP TEACHING AND CREDIT ASSIGNMENT

On LIBERO-Object, we vary bandwidth, temperature, and teacher assignment one factor at a time, holding the remaining settings fixed. Figure 5 and Table 5 summarize all tested settings.

Temporal window. $k = 30$ gives the best AUC and TTS@95. Although $k = 50$ has the highest Early SR, neither $k = 1$ nor $k = 50$ reaches 95% within its budget, illustrating the trade-off between execution-speed tolerance and phase mixing.

Credit temperature. Temperature controls concentration of negative credit. Exponential annealing gives the best AUC; linear annealing reaches 95% three steps earlier but has lower Early SR (77.31 vs. 85.64).

Teacher assignment. ST uses one teacher per group, MTA averages successes, and NNT selects a teacher per failure (Appendix A.3). NNT leads all metrics; ST requires 81 steps and MTA never reaches 95%, supporting failure-specific teaching. Successful rollouts retain uniform weights.

Taken together, the three analyses indicate that GTPO benefits from a moderate temporal context, smooth credit concentration, and failure-specific teacher selection. These choices improve both the speed of reaching high success and the area under the learning curve, so we use the default setting ($k = 30$, $\tau = \text{Exp.}$, NNT) in all subsequent experiments.

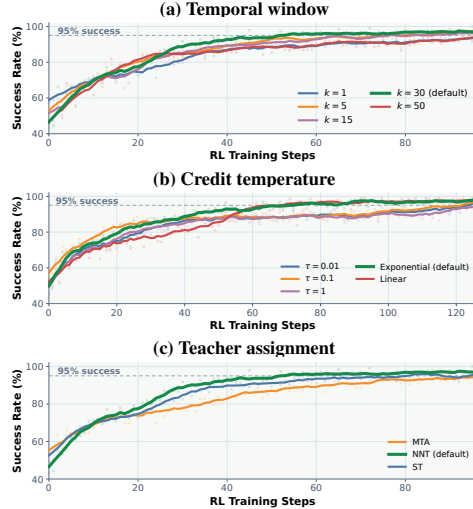


Table 5: Design-analysis metrics on LIBERO-Object. Exp. and Linear anneal τ from 0.05 to 1; $>$ denotes a threshold not reached within that analysis horizon.

Analysis	Setting	AUC	Early	TTS@95
Window k	1	84.02	77.83	> 96
	5	86.91	83.63	75
	15	85.95	82.07	77
	30	87.96	83.04	52
	50	84.25	84.75	> 96
Temperature τ	0.01	84.93	82.81	124
	0.1	86.90	85.83	122
	1	84.37	82.85	> 126
	Exp.	89.38	85.64	71
	Linear	87.11	77.31	68
Teacher	ST	85.99	78.91	81
	MTA	83.13	76.04	> 96
	NNT	87.96	83.04	52

Figure 5: Design analyses on LIBERO-Object; three efficiency curves are stacked in panels (a)–(c) and the complete metric table is shown at right.

Visualization of the OT-credit mechanism. Figure 6 illustrates a successful teacher and a failed student. Temporal reweighting applies to the student; successful rollouts retain uniform weights.

The failed rollout (bottom) contains an empty-space grasp marked in red. The weight profile illustrates how visual divergence from the teacher can concentrate negative advantage near an annotated error. This qualitative example explains the intended allocation mechanism; it does not establish that visual distance identifies the causal error or validate localization accuracy. Changes in appearance or delayed action effects can also affect the weights.



Figure 6: Illustration of OT credit allocation for a successful teacher (top) and failed student (bottom). The weight profile refers to the student; red marks the annotated failed grasp.

5 REAL-WORLD EXPERIMENTS

We evaluate five physical tasks: block stacking, battery insertion, component soldering, screw driving, and bottle-cap removal, using PPO-style on-policy updates with intervention guidance as detailed in Appendix C.

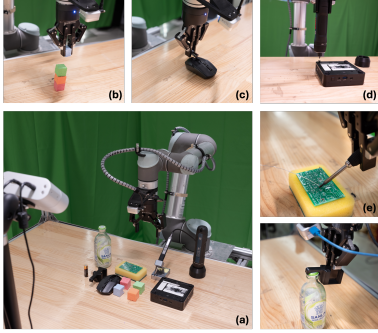


Figure 7: Real-world setup and tasks: (a) scene; (b) stack blocks; (c) insert battery; (d) solder component; (e) drive screw; and (f) pop the top.

Setup and training protocol. We deploy $\pi_{0.5}$ (Black et al., 2025) on a UR5 with base and wrist cameras. A checkpoint fine-tuned on 50 demonstrations initializes GRPO/GTPO for three on-policy rounds. Both use a PPO-style clipped probability-ratio loss: GRPO uses group-relative advantages, while GTPO applies temporal OT weights. Round one shares rollouts; later rounds collect fresh data with each method’s latest policy. Groups use nearby same-task initial states and require an autonomous success. Operators intervene near errors; interrupted prefixes count as failures, and corrective segments enter only an auxiliary SFT loss. No critic or GAE is used.

Results. Each point uses 20 physical trials under shared poses and binary success criteria. Figure 8 shows two representative trajectories and final success on all tasks. GTPO reaches 85–95% after round three and improves over GRPO by 10–20 percentage points, with the largest gains on soldering and screw driving under the same initialization and takeover protocol.

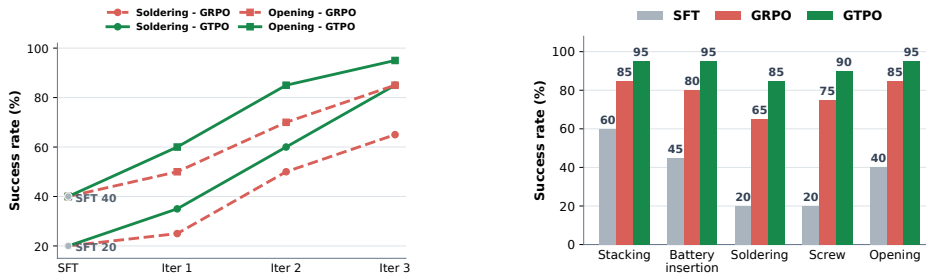


Figure 8: Real-world success from 20 trials per point. (a) SFT initialization and three on-policy training rounds for two representative tasks; solid/dashed lines denote GTPO/GRPO. (b) SFT initialization and final-iteration success on all five tasks.

6 CONCLUSION AND LIMITATIONS

We introduced GTPO, which converts group-relative feedback into chunk-level credit by pairing failures with in-group successful teachers and aligning their visual trajectories with temporally masked OT. Across simulation and physical tasks, it improves success and efficiency without a separately trained process-reward model. **Limitations.** Real-world reinforcement learning remains inefficient under costly, safety-constrained interaction. Visual trajectory comparisons can misestimate GTPO weights under occlusion, viewpoint changes, or visually similar but dynamically different actions.

AI use statement. We used generative AI for language editing, presentation feedback, and figure/layout refinement. The authors produced and verified all method, data, analyses, and final content against implementation records.

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A METHOD DETAILS

A.1 OPTIMAL TRANSPORT BACKGROUND

Optimal Transport (OT) (Peyré & Cuturi, 2019) compares probability distributions by finding the minimum-cost coupling between them. Given discrete marginals $\mu \in \Delta^n$, $\nu \in \Delta^m$ and a non-negative cost matrix $\mathcal{C} \in \mathbb{R}_{\geq 0}^{n \times m}$, the Kantorovich formulation seeks a transport plan $\mathcal{T} \in \mathbb{R}_{\geq 0}^{n \times m}$ minimizing

$$\mathcal{T}^* = \operatorname{argmin}_{\mathcal{T} \in \mathbb{R}_{\geq 0}^{n \times m}} \langle \mathcal{T}, \mathcal{C} \rangle \quad \text{s.t.} \quad \mathcal{T} \mathbf{1} = \mu, \quad \mathcal{T}^\top \mathbf{1} = \nu, \quad (16)$$

where entry $\mathcal{T}_{ab}$ is the probability mass moved from source a to target b , and $\langle \cdot, \cdot \rangle$ is the Frobenius inner product. The minimum objective value is the Wasserstein distance between μ and ν under $\mathcal{C}$. Solving (16) exactly requires a linear program, which is expensive for long sequences. The Sinkhorn algorithm (Cuturi, 2013) adds an entropic regularizer $H(\mathcal{T}) = -\sum_{a,b} \mathcal{T}_{ab} \log \mathcal{T}_{ab}$ for a smooth, efficient relaxation:

$$\mathcal{T}_\varepsilon^* = \operatorname{argmin}_{\mathcal{T} \geq 0} \langle \mathcal{T}, \mathcal{C} \rangle - \varepsilon H(\mathcal{T}), \quad \text{s.t.} \quad \mathcal{T} \mathbf{1} = \mu, \quad \mathcal{T}^\top \mathbf{1} = \nu, \quad (17)$$

whose solution decomposes as $\mathcal{T}_\varepsilon^* = \operatorname{diag}(u) \mathcal{K} \operatorname{diag}(v)$ with kernel $\mathcal{K}_{ab} = \exp(-\mathcal{C}_{ab}/\varepsilon)$, where u, v are found by alternating row/column normalization in log-space. A key limitation of (17) is that it is *permutation-invariant*: shuffling the source or target sequence leaves the optimal cost unchanged, so temporal ordering must be imposed separately (Section 3), which is essential for sequential tasks such as robotic manipulation.

A.2 TEMPORALLY MASKED SINKHORN DERIVATION

We derive the solution of (10) for one trajectory pair and omit the superscript (i, j) for clarity. Let $\mathcal{I} = \{(a, b) : \mathcal{M}_{ab} = 1\}$ be the set of permitted matches. Using $-H(\mathcal{T}) = \sum_{a,b} \mathcal{T}_{ab} \log \mathcal{T}_{ab}$, and noting that $\sum_{a,b} \mathcal{T}_{ab} = 1$, we may write the objective equivalently as

$$\begin{aligned} \min_{\mathcal{T} \geq 0} \quad & \sum_{(a,b) \in \mathcal{I}} [\mathcal{T}_{ab} \mathcal{C}_{ab} + \varepsilon \mathcal{T}_{ab} (\log \mathcal{T}_{ab} - 1)] \\ \text{s.t.} \quad & \mathcal{T} \mathbf{1} = \mu, \quad \mathcal{T}^\top \mathbf{1} = \nu, \quad \mathcal{T}_{ab} = 0 \quad \forall (a, b) \notin \mathcal{I}. \end{aligned} \quad (18)$$

The subtraction of 1 inside the entropy term changes the objective only by the constant $-\varepsilon$ and therefore does not change its minimizer.

Lagrangian and stationary point. Introduce multipliers $\alpha \in \mathbb{R}^n$ and $\beta \in \mathbb{R}^m$ for the row and column marginals. Restricting optimization to the permitted support $\mathcal{I}$, the Lagrangian is

$$\begin{aligned} \mathfrak{L}(\mathcal{T}, \alpha, \beta) = & \sum_{(a,b) \in \mathcal{I}} [\mathcal{T}_{ab} \mathcal{C}_{ab} + \varepsilon \mathcal{T}_{ab} (\log \mathcal{T}_{ab} - 1)] \\ & + \sum_a \alpha_a \left(\mu_a - \sum_{b: (a,b) \in \mathcal{I}} \mathcal{T}_{ab} \right) + \sum_b \beta_b \left(\nu_b - \sum_{a: (a,b) \in \mathcal{I}} \mathcal{T}_{ab} \right). \end{aligned} \quad (19)$$

For every permitted entry, the stationarity condition is

$$\frac{\partial \mathfrak{L}}{\partial \mathcal{T}_{ab}} = \mathcal{C}_{ab} + \varepsilon \log \mathcal{T}_{ab} - \alpha_a - \beta_b = 0, \quad (20)$$

which gives

$$\mathcal{T}_{ab} = \exp\left(\frac{\alpha_a + \beta_b - \mathcal{C}_{ab}}{\varepsilon}\right) = u_a \mathcal{K}_{ab} v_b. \quad (21)$$

Here $u_a = \exp(\alpha_a/\varepsilon)$, $v_b = \exp(\beta_b/\varepsilon)$, and the masked Gibbs kernel is

$$\mathcal{K}_{ab} = \mathcal{M}_{ab} \exp(-\mathcal{C}_{ab}/\varepsilon). \quad (22)$$

Thus forbidden entries are exactly zero, while the complete solution has the matrix form

$$\mathcal{T}^* = \text{diag}(u) \mathcal{K} \text{diag}(v). \quad (23)$$

Sinkhorn scaling. Substituting (23) into the marginal constraints yields

$$u \odot (\mathcal{K}v) = \mu, \quad v \odot (\mathcal{K}^\top u) = \nu, \quad (24)$$

where $\odot$ denotes element-wise multiplication. Therefore, u and v can be obtained by alternating updates

$$u^{(q+1)} = \frac{\mu}{\mathcal{K}v^{(q)}}, \quad v^{(q+1)} = \frac{\nu}{\mathcal{K}^\top u^{(q+1)}}, \quad (25)$$

where division is element-wise. Starting from $v^{(0)} = \mathbf{1}$, these updates are repeated until both marginal residuals $\|\mathcal{T}\mathbf{1} - \mu\|_1$ and $\|\mathcal{T}^\top \mathbf{1} - \nu\|_1$ fall below a tolerance.

Log-domain implementation. Direct evaluation of $\exp(-\mathcal{C}/\varepsilon)$ may underflow when ε is small. We instead define dual potentials $f = \varepsilon \log u$ and $g = \varepsilon \log v$ and perform the updates as

$$f_a = \varepsilon \log \mu_a - \varepsilon \log \sum_{b: \mathcal{M}_{ab}=1} \left(\frac{g_b - \mathcal{C}_{ab}}{\varepsilon} \right), \quad (26)$$

$$g_b = \varepsilon \log \nu_b - \varepsilon \log \sum_{a: \mathcal{M}_{ab}=1} \left(\frac{f_a - \mathcal{C}_{ab}}{\varepsilon} \right). \quad (27)$$

After convergence, the transport plan is recovered by

$$\mathcal{T}_{ab}^* = \begin{cases} \exp((f_a + g_b - \mathcal{C}_{ab})/\varepsilon), & \mathcal{M}_{ab} = 1, \\ 0, & \mathcal{M}_{ab} = 0. \end{cases} \quad (28)$$

The temporal mask must contain at least one permitted entry in every row and column and admit a coupling with the specified marginals; we choose k accordingly. Each dense iteration costs $O(nm)$ time, or $O(|\mathcal{I}|)$ when the banded support is stored sparsely. Finally, the trajectory distance used for teacher selection is the unregularized transport cost $\langle \mathcal{T}^*, \mathcal{C} \rangle$, as defined in (11); entropy is used only to obtain a stable alignment.

A.3 TEACHER ASSIGNMENT STRATEGIES

We formalize the three teacher assignments compared in Section 4.4. Recall that $\mathcal{S}$ and $\mathcal{F}$ are the successful and failed rollouts in one GRPO group. We exclude rollouts with fewer than two valid chunks from alignment; if no valid teacher remains, weights are uniform. Below, $\mathcal{S}$ denotes the remaining teacher pool. For each valid pair (i, j) , temporally masked OT on shifted features of lengths $L_i = T_i - 1$ and $L_j = T_j - 1$ produces the plan $\mathcal{T}^{*(i,j)}$, cost matrix $\mathcal{C}^{(i,j)}$, and trajectory distance $d_{ij} = d_{\text{traj}}(\mathcal{X}_i, \mathcal{X}_j)$. To make the dependence on teacher j explicit, define the pair-specific alignment score and normalized credit weight as

$$s_{i,t}^{(j)} = - \sum_{b=0}^{L_j-1} \mathcal{T}_{tb}^{*(i,j)} \mathcal{C}_{tb}^{(i,j)}, \quad 0 \leq t < L_i, \quad (29)$$

$$\tilde{w}_{i,t}^{(j)} = L_i \frac{\exp(-s_{i,t}^{(j)}/\tau)}{\sum_{u=0}^{L_i-1} \exp(-s_{i,u}^{(j)}/\tau)}, \quad \sum_{t=0}^{L_i-1} \tilde{w}_{i,t}^{(j)} = L_i. \quad (30)$$

Set $w_{i,t}^{(j)} = \tilde{w}_{i,t}^{(j)}$ for $t < L_i$ and $w_{i,T_i-1}^{(j)} = 1$. Thus $\sum_{t=0}^{T_i-1} w_{i,t}^{(j)} = T_i$. Higher transport cost gives a larger penalty weight for negative-advantage rollouts, consistently with Equation (14).

Single teacher (ST). ST chooses one successful rollout for the entire group. Let

$$j_{\text{ST}}^* \in \operatorname{argmax}_{j \in \mathcal{S}} r_j, \quad w_{i,t}^{\text{ST}} = w_{i,t}^{(j_{\text{ST}}^*)}, \quad i \in \mathcal{F}. \quad (31)$$

The same j_{ST}^* supervises every failure. When several successful rollouts have the same terminal reward, as in our binary-reward setting, one is selected once with a fixed seeded tie break and then shared by the group. ST is inexpensive, but a single teacher may be visually or behaviorally mismatched to some failures.

Multi-teacher averaging (MTA). MTA aligns each failed rollout with every successful rollout and averages the resulting teacher-specific credit weights:

$$w_{i,t}^{\text{MTA}} = \frac{1}{|\mathcal{S}|} \sum_{j \in \mathcal{S}} w_{i,t}^{(j)}, \quad i \in \mathcal{F}. \quad (32)$$

Thus MTA averages normalized temporal credits rather than raw feature trajectories or OT cost matrices. It uses all available successes, but irrelevant teachers can dilute the sharp temporal signal from a well-matched demonstration.

Nearest-neighbor teacher (NNT). NNT assigns a different teacher to each failure by minimizing the pairwise trajectory distance:

$$j_i^* = \operatorname{argmin}_{j \in \mathcal{S}} d_{ij}, \quad w_{i,t}^{\text{NNT}} = w_{i,t}^{(j_i^*)}, \quad i \in \mathcal{F}. \quad (33)$$

Unlike ST, NNT adapts the teacher to the current failure; unlike MTA, it avoids averaging incompatible behaviors. This is the default assignment used by GTPO.

For any assignment $q \in \{\text{ST}, \text{MTA}, \text{NNT}\}$, the final chunk-level advantage is

$$\hat{A}_{i,t}^q = A_i w_{i,t}^q m_{i,t}. \quad (34)$$

All three variants preserve $\sum_t w_{i,t}^q = T_i$, so they change only the temporal distribution of the original trajectory advantage. Successful rollouts, and all rollouts in a group with $\mathcal{S} = \emptyset$, retain $w_{i,t} = 1$.

A.4 GTPO ALGORITHM

Algorithm 1 GTPO: Group Teaching via Temporal Optimal Transport

Require: Scores $\{r_i\}_{i=1}^G$, shifted features $\mathcal{X}_i = (x_{i,1}, \dots, x_{i,T_i-1})$, valid lengths T_i , masks $m_{i,t}$, bandwidth k , temperature τ , entropic weight ε

Ensure: Dense advantages $\{\hat{A}_{i,t}\}$

1: $A_i \leftarrow (r_i - \mu_g) / (\sigma_g + \varepsilon_{\text{std}})$ for all $i \in \{1, \dots, G\}$ ▷ Eq. (5)

2: $\mathcal{S} \leftarrow \{j : r_j > 0\}$, $\mathcal{F} \leftarrow \{i : r_i < 0\}$

3: $L_i \leftarrow T_i - 1$ for all i ; $\mathcal{S}_{\text{valid}} \leftarrow \{j \in \mathcal{S} : T_j \geq 2\}$

4: **for** $i = 1, \dots, G$ **do**

5: $w_{i,t} \leftarrow 1$ for all valid t ▷ Also covers short rollouts

6: **if** $i \in \mathcal{F}$ and $T_i \geq 2$ and $\mathcal{S}_{\text{valid}} \neq \emptyset$ **then**

7: $j_i^* \leftarrow \operatorname{argmin}_{j \in \mathcal{S}_{\text{valid}}} \text{SinkhornCost}(\mathcal{X}_i, \mathcal{X}_j; k, \varepsilon)$ ▷ Eq. (12)

8: $(\mathcal{C}, \mathcal{M}, \mathcal{T}^*) \leftarrow \text{MaskedOT}(\mathcal{X}_i, \mathcal{X}_{j_i^*}; k, \varepsilon)$

9: $s_{i,t} \leftarrow -\sum_{b=0}^{L_{j_i^*}-1} \mathcal{T}_{tb}^* \mathcal{C}_{tb}$ for $0 \leq t < L_i$ ▷ Eq. (13)

10: $\tilde{w}_i \leftarrow L_i \text{softmax}(-s_i / \tau)$ ▷ Eq. (14)

11: $w_{i,t} \leftarrow \tilde{w}_{i,t}$ for $0 \leq t < L_i$; $w_{i,T_i-1} \leftarrow 1$

12: **end if**

13: $\hat{A}_{i,t} \leftarrow A_i \cdot w_{i,t} \cdot m_{i,t}$ for all t ▷ Eq. (15)

14: **end for**

15: **return** $\{\hat{A}_{i,t}\}$

B SIMULATION EXPERIMENT DETAILS

B.1 BENCHMARK AND POLICY INITIALIZATION

The configuration below corresponds to the LIBERO-Object experiment with OpenVLA-OFT and GRPO-OT. The suite contains ten object-manipulation tasks. We initialize the policy from the LIBERO-130 LoRA SFT checkpoint, which is trained jointly on all 130 LIBERO tasks rather than on an Object-specific single-trajectory subset. OpenVLA-OFT has approximately 8B parameters and combines a Llama-2-7B language backbone with fused DINOv2 and SigLIP visual encoders. Reinforcement learning updates all policy parameters (LoRA is disabled), without a value head or a critic.

Table 6: Simulation model and input configuration.

Parameter	Setting
Policy camera	One third-person camera
Image resolution	256×256 rendering; 224×224 center-cropped policy input
Wrist camera	Rendered by the environment; not provided to the policy
Action space	7D (translation, rotation, and gripper)
Action-chunk horizon	8
Action discretization	256 bins
Action statistics	LIBERO-130 SFT normalization statistics
Hidden / vocabulary size	4,096 / 32,000
Proprioception / FiLM	Both disabled
Maximum prompt length	128
Maximum generated length	1,024

B.2 TRAINING AND INFERENCE CONFIGURATION

Each policy iteration collects rollouts from groups of matched initial states and then updates the actor. Episodes contain at most 320 environment steps, corresponding to at most 40 policy decisions with an action-chunk horizon of 8. Advantages are normalized within each group. Table 7 gives the rollout and optimization settings.

Table 7: Simulation rollout and policy optimization.

Parameter	Setting
Training budget	At most 1,000 policy iterations
Rollout collection	8 rollout epochs per iteration; 128 parallel environments
Group size	$G = 8$
Optimizer	Adam; $(\beta_1, \beta_2) = (0.9, 0.999)$, $\epsilon = 10^{-8}$
Learning rate	2×10^{-5}
Weight decay	0.01
Gradient clipping	Global norm of 1.0
Batch size	10,240 globally; micro-batch size 32 per GPU
Optimization epochs	One per collected batch
Log probabilities / entropy	Token-level
Loss aggregation	Token-mean
Clipping	Lower/upper ratios 0.20/0.28; dual-clip coefficient 3.0
KL / entropy coefficients	Both zero
Reward processing	Action-level rewards, scaled by 5.0
Reward filtering	Enabled; lower/upper bounds 0.5/4.5
Sampling	Stochastic; temperature 1.6, top- $p = 1$, no top- k truncation, repetition penalty 1.0

We compute cosine feature costs without temporal subsampling. The temporal OT settings for this configuration are listed in Table 8.

Table 8: Temporal optimal-transport configuration.

Parameter	Setting
Feature cost	Cosine distance
Temporal subsampling	None
Mask half-bandwidth	$k = 15$
Sinkhorn regularization	$\varepsilon = 0.01$
Sinkhorn iterations	100
Credit temperature	Exponential schedule from 0.01 to 1.0 over the first 80 policy iterations

Evaluation uses the same stochastic decoding parameters as training. Table 9 summarizes evaluation and systems settings. Training uses a single node equipped with NVIDIA A800 GPUs.

Table 9: Simulation evaluation and systems configuration.

Parameter	Setting
Evaluation frequency	Every policy iteration
Evaluation environments	192 parallel environments with automatic resets
Evaluation rollouts	One rollout per fixed reset-state identifier
Episode horizon	320 environment steps, for both training and evaluation
Evaluation sampling	Temperature 1.6, top- $p = 1$, no top- k truncation, repetition penalty 1.0
Hardware	Single node with NVIDIA A800 GPUs
Training backend	Fully sharded data parallelism
Memory settings	Gradient checkpointing and actor/rollout offloading
Precision	bfloat16
Actor / environment seed	1234 / 0
Checkpoint interval	Every 500 policy iterations

B.3 COSINE DISTANCE IN THE OT ABLATION

Section 4.3 compares two ways of matching visual trajectories, both using cosine distance. GTPO-pointwise retains nearest-success teaching and normalized chunk weights but replaces masked Sinkhorn alignment with pointwise matching. Thus this comparison tests the alignment mechanism, rather than replacing cosine distance with a different feature metric.

Feature distance. For a failed rollout i and a successful teacher candidate j , we use the shifted features $\mathcal{X}_i = (x_{i,1}, \dots, x_{i,T_i-1})$ and $\mathcal{X}_j = (x_{j,1}, \dots, x_{j,T_j-1})$ from Equation (7). The feature following an action chunk represents its observed consequence. For valid indices $0 \leq a < L_i$ and $0 \leq b < L_j$, the cost is

$$c_{ab}^{(i,j)} = 1 - \frac{x_{i,a+1}^\top x_{j,b+1}}{\|x_{i,a+1}\|_2 \|x_{j,b+1}\|_2 + 10^{-8}}. \quad (35)$$

The computation uses float32 features and the stabilizer 10^{-8} in the denominator. For nonzero features, costs are approximately 0 for the same direction, 1 for orthogonal directions, and 2 for opposite directions. This measures visual dissimilarity, not task success or an action’s physical correctness directly.

Pointwise matching versus OT. For equal-length trajectories indexed on the same time grid, pointwise comparison reads $c_{tt}^{(i,j)}$, comparing the two successor observations at each corresponding chunk. It does not search for an alternative match when one rollout executes a sub-action earlier or later. In contrast, GTPO uses the full pairwise matrix within the temporal mask and computes

$$s_{i,t}^{\text{OT}} = - \sum_{b=0}^{L_j-1} \mathcal{T}_{tb}^{\star(i,j)} c_{tb}^{(i,j)}. \quad (36)$$

The transport plan can distribute a chunk’s mass across nearby teacher chunks while satisfying both trajectory marginals. Consequently, a timing offset can increase a pointwise distance even when the two rollouts exhibit similar behavior, whereas OT can accommodate that offset within the permitted temporal band.

From scores to penalty weights. The shared credit transformation takes alignment scores $s_{i,t}$, with larger scores denoting closer matches, and applies

$$\tilde{w}_{i,t} = L_i \frac{\exp(-s_{i,t}/\tau)}{\sum_{u=0}^{L_i-1} \exp(-s_{i,u}/\tau)}, \quad \hat{A}_{i,t} = A_i w_{i,t} m_{i,t}. \quad (37)$$

Here $w_{i,t} = \tilde{w}_{i,t}$ for $t < L_i$, and the final chunk retains weight 1 because it has no stored successor feature. A larger mismatch yields a lower score and a larger weight; for a failed rollout with $A_i < 0$, this strengthens the negative advantage. Normalization preserves $\sum_t w_{i,t} = T_i$. Successful rollouts and failures without a valid teacher retain uniform weights. Score scale matters: the OT row cost in Equation (36) includes transport mass $1/L_i$, so replacing it by a raw cosine distance changes the effective temperature unless that scale is accounted for.

B.4 EVALUATION METRICS

Efficiency metrics. We report normalized area under the success-rate curve over the common per-suite horizon (AUC-SR), success at 25% of that horizon (Early SR), and the first step reaching 95% success (TTS@95). We apply the seven-checkpoint moving average used in Figure 3, compute each metric per suite, and macro-average across the four suites. If a run never reaches 95%, TTS@95 is right-censored at its final training step; substituting that budget yields only a lower bound on GTPO’s efficiency gain.

OT ablation metrics. For the comparison in Section 4.3, we apply a centered seven-checkpoint moving average to success rates, replicating boundary values at the endpoints. Normalized AUC-SR is the mean of the smoothed success rates; Early SR is the smoothed success rate at 25% of training, and TTS@95 is the first recorded step at which the smoothed success rate reaches 95%. Final SR is the final raw evaluation. These are descriptive statistics of one exported series per method; the export does not provide independent seed-level variation.

Reward diagnostics. We measure reward–success separation $\mathbb{E}[R \mid \text{success}] - \mathbb{E}[R \mid \text{failure}]$, return-based success AUROC, Pearson/Spearman return–success correlation, and the rates of high-reward failures and low-reward successes. The latter use the 25th percentile of successful returns and the 75th percentile of failed returns, respectively.

B.5 DESIGN-ANALYSIS PROTOCOL

On LIBERO-Object, we vary one factor at a time while holding the remaining settings at their defaults: temporal window $k \in \{1, 5, 15, 30, 50\}$; constant $\tau \in \{0.01, 0.1, 1\}$ or linear/exponential $0.05 \rightarrow 1$ schedules; and ST, MTA, or NNT teacher assignment. Efficiency metrics are computed from the seven-checkpoint moving averages over the common training horizon within each analysis.

C REAL-WORLD EXPERIMENT DETAILS

C.1 BACKBONE AND SFT CONFIGURATION

All real-world experiments use $\pi_{0.5}$ (Black et al., 2025) as the VLA backbone. For each task, we collect 50 demonstrations and fine-tune a task-specific SFT checkpoint. The same data and checkpoint initialize GRPO and GTPO. The reported SFT baseline is this checkpoint before takeover training. Initial SFT uses eight GPUs with the following settings:

Table 10: Real-world SFT and inference configuration.

Parameter	Setting
Optimizer	AdamW; $(\beta_1, \beta_2) = (0.9, 0.95)$, $\varepsilon = 10^{-8}$
Weight decay	10^{-10}
Learning-rate schedule	Linear warmup for 1,000 steps to 2.5×10^{-5} , then cosine decay to 2.5×10^{-6}
Batch size	512 globally; 64 per GPU
Training steps	30,000
Checkpoint interval	Every 2,500 steps
Gradient clipping	Global norm of 1.0
Precision	bfloat16
Visual input	Base and wrist cameras at 224×224
Action space	Physical 7D TCP actions (xyz, rotnvec, gripper), padded to 32 dimensions
Action horizon	50
Action convention	TCP pose actions are deltas; gripper command is absolute
SFT flow-matching training	$u \sim \text{Beta}(1.5, 1.0)$
Inference	10 denoising steps

C.2 THREE-ROUND ON-POLICY COLLECTION WITH TAKEOVER

Starting from the common SFT checkpoint, GRPO and GTPO perform three rounds of robot inference, operator takeover, and policy update. During collection, the current policy controls the robot until it is about to make an error or an error has already occurred; only then does the operator take over and provide a corrective continuation. The policy-generated prefix preceding the takeover is terminated and labeled as a failed sample, irrespective of whether operator control later completes the task. This prevents a human recovery from being attributed to the policy. We split each round into two disjoint streams. The policy stream $\mathcal{D}_{\text{pol},M}^{(q)}$ contains autonomous rollouts and the policy-generated prefixes before takeover; each such prefix receives a failure label. The takeover stream $\mathcal{D}_{\text{to},M}^{(q)}$ contains only the operator-controlled corrective continuation. Both streams record the instruction, image observations, executed action chunks, initial robot state, intervention indicator, and binary label. All methods use identical reset, takeover, and success criteria, as well as the same rollout and update budgets in every round. For policy-generated chunks, the rollout buffer additionally stores the collecting policy version, its log probabilities, and the sampling variables needed to recompute those log probabilities during optimization.

Let $M \in \{\text{GRPO}, \text{GTPO}\}$ index the training branch and $\theta_M^{(q)}$ its parameters after round q . Because all branches have identical parameters before round one, we collect one common pair of policy and takeover streams and use it to update every method. In rounds two and three, each method deploys its own latest checkpoint and trains only on the two streams generated during its own rollout:

$$(\mathcal{D}_{\text{pol},M}^{(1)}, \mathcal{D}_{\text{to},M}^{(1)}) = (\mathcal{D}_{\text{pol}}^{(1)}, \mathcal{D}_{\text{to}}^{(1)}), \quad (38)$$

$$(\mathcal{D}_{\text{pol},M}^{(q)}, \mathcal{D}_{\text{to},M}^{(q)}) \sim \text{RolloutTakeover}\left(\pi_{\theta_M^{(q-1)}}\right), \quad q \in \{2, 3\}. \quad (39)$$

The shared first round controls for collection variability. Each round freezes the behavior policy at $\theta_{\text{old},M}^{(q)} = \theta_M^{(q-1)}$ for collection and holds its recorded log probabilities fixed during the ensuing clipped update. Subsequent rounds collect fresh rollouts with each updated policy, without replaying earlier policy rollouts. Operator actions in $\mathcal{D}_{\text{to},M}^{(q)}$ provide separate auxiliary SFT supervision.

C.3 ROLLOUT GROUPS AND TEMPORAL TEACHING

Round-wise rollout groups. Groups are constructed exclusively from the policy stream $\mathcal{D}_{\text{pol},M}^{(q)}$. Because exact repeated resets are difficult on a physical robot, a group contains same-task rollouts with nearby initial visual features and proprioceptive states. Groups without an autonomous success are skipped; within each valid group, binary outcomes are normalized to obtain the relative advantages A_i in Equation (5). Takeover segments never enter group normalization and can never be selected as teachers.

Temporal group teaching. Teacher selection and credit assignment remain identical to the simulation method. Each failed rollout selects its nearest successful trajectory according to Equation (12); the temporally masked Sinkhorn plan then produces $w_{i,t}$ and $\hat{A}_{i,t} = A_i w_{i,t} m_{i,t}$. Thus, each round uses successful behavior generated under the current policy as its teaching signal while retaining temporal OT credit allocation.

C.4 PPO-STYLE CLIPPED POLICY UPDATE

Policy ratios. Both branches use PPO-style clipping (Schulman et al., 2017) with group-relative advantages, without a critic or GAE. Let $s_{\theta,i,t}$ be the stochastic $\pi_{0.5}$ interface’s log-probability score, recomputed on the stored observations and denoising samples using the same masking and aggregation as collection. Suppressing within-chunk component indices, the ratio is

$$\rho_{i,t}^{(q)}(\theta) = \exp\left(s_{\theta,i,t} - \text{stopgrad}\left(s_{\theta_{\text{old},M}^{(q)},i,t}\right)\right). \quad (40)$$

Scores come from stochastic sampling likelihoods, not flow-matching regression errors.

Group-relative clipped objective. The two branches differ in the advantage supplied to the same surrogate:

$$B_{i,t}^{\text{GRPO}} = A_i, \quad B_{i,t}^{\text{GTPO}} = \hat{A}_{i,t}. \quad (41)$$

Group advantages and OT weights are computed from the collected round and held fixed during optimization. Write $\bar{\rho}_{i,t}^{(q)} = \text{clip}(\rho_{i,t}^{(q)}, 1 - \varepsilon_{\text{clip}}, 1 + \varepsilon_{\text{clip}})$. The policy loss is

$$\mathcal{L}_{\text{clip},M}^{(q)}(\theta) = -\frac{1}{Z_{\text{pol}}} \sum_{(i,t) \in \mathcal{D}_{\text{pol},M}^{(q)}} m_{i,t} \min\left(\rho_{i,t}^{(q)} B_{i,t}^M, \bar{\rho}_{i,t}^{(q)} B_{i,t}^M\right), \quad (42)$$

where Z_{pol} counts valid policy loss terms. Signed chunk advantages are broadcast to the corresponding probability components and enter the clipped objective directly.

Auxiliary takeover supervision. Only operator-controlled corrective segments receive the SFT flow-matching loss. For action chunk $a_{i,t}$, Gaussian noise z , and flow time $u \sim \text{Beta}(1.5, 1.0)$, let $a_{i,t}^u = (1 - u)z + ua_{i,t}$ and $v_{i,t}^* = a_{i,t} - z$. Then

$$\mathcal{L}_{\text{to-SFT}}^{(q)}(\theta) = \frac{1}{Z_{\text{to}}} \sum_{(i,t) \in \mathcal{D}_{\text{to},M}^{(q)}} m_{i,t} \|v_{\theta}(a_{i,t}^u, o_{i,t}, \ell, u) - v_{i,t}^*\|_2^2, \quad (43)$$

where Z_{to} is the number of valid takeover chunks. The full objective for each branch is

$$\mathcal{L}_M^{(q)}(\theta) = \mathcal{L}_{\text{clip},M}^{(q)}(\theta) + \lambda_{\text{SFT}} \mathcal{L}_{\text{to-SFT}}^{(q)}(\theta), \quad (44)$$

with the same auxiliary coefficient for both branches. Operator-controlled actions enter neither the probability-ratio objective nor group-relative or temporal-OT advantage estimation.

C.5 BASELINES AND EVALUATION

SFT denotes the common initialization before the three rounds. GRPO and GTPO share the clipped objective and auxiliary takeover SFT loss, differing only in the advantages in Equation (41). We evaluate SFT and each round’s checkpoints without operator takeover, using the same pose protocol and binary success criteria with 20 trials per task and checkpoint.